

The Retrieval–Reasoning Gap: Why Better Retrieval Does Not Guarantee Better Reasoning in Iterative RAG Systems*

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April 2026

Abstract

Standard RAG evaluation assumes that improving retrieval quality directly improves end-task accuracy—an assumption we show fails for multi-hop reasoning. We identify the *retrieval–reasoning gap*: retrieval-side metric gains do not transfer to reasoning accuracy when the language model is held constant. Using a six-configuration ablation on 65 multi-hop queries (synthetic telecom logs, 1,115 entries), we find Precision@K varies by 0.24 ($p < 0.001$) while root-cause accuracy remains statistically flat ($\Delta = 0.05$, $p = 0.08$, n.s.; Pearson $r = 0.244$). A multi-LLM comparison (8B–70B) with fixed retrieval confirms reasoning is independently bounded by model capability. We propose a composite metric (semantic similarity + ROUGE-L + structured scoring) that captures diagnostic quality beyond keyword overlap ($r = 0.849$ with semantic evaluation). These findings generalize beyond text-only pipelines: any multimodal retrieval system that assembles heterogeneous evidence before generation faces the same structural gap between retrieval and reasoning.

1 Introduction

Evaluation of retrieval-augmented generation (RAG) systems [Lewis et al., 2020, Gao et al., 2023] typically reports two metric families side by side: retrieval quality (Precision@K, Recall@K, nDCG) and end-task quality (answer accuracy, F1). The implicit assumption is that improving retrieval improves the end-task. For multi-hop RAG—where evidence must be assembled from multiple sources before a language model reasons over it—this assumption is load-bearing yet under-examined. This work challenges that core assumption with controlled empirical evidence: improving retrieval alone is insufficient for multi-hop reasoning when the reader model is fixed.

This paper presents the **retrieval–reasoning gap**: the observation that improvements in retrieval-side metrics do not translate to improvements in reasoning accuracy, and that this gap is a structural property of multi-hop RAG rather than an artifact of dataset size or domain.

Our experimental design isolates retrieval from reasoning through three mechanisms:

1. A **six-configuration ablation study** that varies the retrieval pipeline ($\pm$ BM25, $\pm$ cross-encoder reranking, $\pm$ iterative refinement) while holding the LLM constant (Section 5.1).
2. A **multi-LLM comparison** that fixes retrieval and varies only the reasoning model across four architectures spanning 8B–70B parameters (Section 5.2).
3. **Improved evaluation metrics**—semantic similarity, structured scoring, composite metric—that capture reasoning quality beyond keyword overlap (Section 5.6).

*Code and data: <https://github.com/yakkalaanugna/Automated-RAG-Agent>

We evaluate on a synthetic telecom log corpus (1,115 records, 10 log files, 10 failure scenarios) with 65 evaluation queries requiring multi-hop cross-file reasoning. The telecom setting is a controlled testbed; the evaluation phenomena we surface—metric disagreement, failure-mode independence, and inadequacy of aggregate scores—generalize to any RAG system performing multi-hop evidence assembly followed by LLM reasoning.

Although our experiments are text-based, the retrieval–reasoning gap is not modality-specific. Multimodal retrieval systems that integrate image+text, video+text, or structured data face the same structural challenge: retrieving relevant heterogeneous evidence does not guarantee that a downstream multimodal generation model will reason correctly over it. Our evaluation framework—decomposing retrieval from reasoning—applies directly to such systems.

Contributions.

1. **Identifying the retrieval–reasoning gap as a structural property.** Across six ablation configurations, Precision@K varies by 0.24 ($p<0.001$) while root-cause accuracy varies by only 0.05 ($p=0.08$, n.s.; Pearson $r=0.244$). Multi-LLM comparison with fixed retrieval shows reasoning accuracy gaps of up to 0.103 attributable to model capability alone.
2. **A decomposition protocol for multi-hop RAG evaluation.** We propose: (a) ablation with fixed LLM, (b) model comparison with fixed retrieval, (c) per-query regime classification (high/low retrieval $\times$ correct/incorrect reasoning)—a framework applicable to both text-only and multimodal pipelines.
3. **Composite evaluation metric.** We show keyword overlap is a noisy proxy and propose a composite score (semantic similarity + ROUGE-L + structured scoring) that better captures diagnostic quality.
4. **Dataset release.** Synthetic telecom log dataset: 1,115 entries, 10 failure scenarios, 65 annotated queries with ground-truth root causes.

2 Related Work

Retrieval-Augmented Generation. Lewis et al. [2020] showed that augmenting LMs with retrieved passages yields large gains on knowledge-intensive tasks; Gao et al. [2023] survey extensions to summarization, code generation, and domain-specific applications. Most RAG systems follow a single-pass architecture insufficient for multi-file log analysis.

Iterative and Multi-Step Retrieval. Self-RAG [Asai et al., 2024] enables self-reflective retrieval; Active RAG [Jiang et al., 2023] uses active learning signals; query rewriting [Wang et al., 2023] reformulates queries after initial passes. These approaches lack principled stopping criteria. Our adaptive confidence-gated iteration addresses this, and our ablation quantifies the contribution of iteration to both retrieval and reasoning.

Hybrid Retrieval and Cross-Encoder Reranking. Fusing BM25 [Robertson and Zaragoza, 2009] and bi-encoder retrieval [Reimers and Gurevych, 2019] is standard practice. Cross-encoder rerankers [Nogueira and Cho, 2019] substantially improve precision; BEIR [Thakur et al., 2021] confirms zero-shot generalization. We show that reranking improves *retrieval* precision but does not proportionally improve *reasoning* accuracy.

RAG Evaluation. RAGAS [Es et al., 2024] and ARES [Saad-Falcon et al., 2024] decompose RAG quality into context-relevance, faithfulness, and answer-relevance. Lost-in-the-middle [Liu et al., 2024] shows LLMs underuse mid-context evidence—invisible to Precision@K. We complement this literature by demonstrating that retrieval and end-task metrics move independently, and that neither alone nor together reveals which failure mode is responsible.

Algorithm 1 Adaptive Iterative RAG

Require: Query q , documents $\mathcal{D}$, max iters M , threshold ϵ

```
1:  $q_1 \leftarrow q$ ;  $c_0 \leftarrow 0$ ; best  $\leftarrow$  NULL; seen  $\leftarrow \emptyset$ 
2: for  $i = 1$  to  $M$  do
3:    $\mathcal{C}_i \leftarrow \text{HYBRIDRETRIEVE}(q_i, \mathcal{D}, N_i)$ 
4:    $\mathcal{R}_i \leftarrow \text{RERANK}(q_i, \mathcal{C}_i, k)$ 
5:    $\mathcal{R}_i \leftarrow \text{DEDUP}(\mathcal{R}_i, \text{seen})$ ; update seen
6:    $a_i \leftarrow \text{LLM}(q_i, \mathcal{R}_i)$ 
7:    $c_i \leftarrow \text{CONFIDENCE}(a_i, \mathcal{R}_i, a_{i-1})$ 
8:   if  $c_i > c_{\text{best}}$  then best  $\leftarrow (a_i, \mathcal{R}_i, c_i)$ 
9:   end if
10:  if  $c_i \geq 0.85$  or drops  $\geq 2$  then break
11:  end if
12:   $q_{i+1} \leftarrow \text{REFINEQUERY}(q, q_i, a_i)$ 
13: end for
14: return best
```

3 Method

3.1 Pipeline Architecture

The system implements a four-stage iterative loop: *Hybrid Retrieval* $\rightarrow$ *Cross-Encoder Reranking* $\rightarrow$ *LLM Reasoning* $\rightarrow$ *Confidence-Gated Iteration*.

Stage 1: Hybrid Candidate Generation. Dense retrieval (all-MiniLM-L6-v2, 384-dim) indexed in ChromaDB fuses with BM25 ($k_1=1.5$, $b=0.75$): $s_{\text{hybrid}}(d) = 0.7 \cdot s_{\text{dense}}(d) + 0.3 \cdot s_{\text{BM25}}(d)$. Over-fetches $N=20\text{--}30$ candidates for reranking.

Stage 2: Cross-Encoder Reranking. ms-marco-MiniLM-L-6-v2 (22M params) rescores each (query, document) pair via joint cross-attention. The cross-encoder score is the *sole* ranking signal, discarding hybrid scores to avoid reintroducing first-stage noise. Top- $k=6$ documents proceed to the LLM.

Stage 3: LLM Analysis. The LLM (default: Llama 3.3 70B, temperature 0) produces structured output: root cause, severity, timeline, reasoning steps, recommendation. Composite confidence combines: mean retrieval score (30%), output completeness (30%), evidence density (20%), inter-iteration consistency (20%).

Stage 4: Confidence-Gated Iteration. Three stopping rules: high-confidence exit (≥ 0.85), plateau detection (2 consecutive non-improvements), hard cap ($M=3$). Returns the *best-scoring* iteration.

3.2 Implementation Details

The retriever uses all-MiniLM-L6-v2 (22M parameters, 384-dim embeddings) indexed in ChromaDB (in-memory). The cross-encoder reranker is ms-marco-MiniLM-L-6-v2 (22M parameters, trained on MS MARCO passages). The default LLM is Llama 3.3 70B served via Groq (temperature 0, max_tokens=2048). Indexing is performed per-session; inference runs on CPU for embedding and reranking, with LLM calls via API. All models are used off-the-shelf without domain fine-tuning.

4 Experimental Setup

4.1 Dataset

We construct a synthetic telecom log corpus for multi-hop retrieval evaluation: **1,115 log entries** across **10 log files** from six network domains (eNodeB, CU-CP, CU-UP, RAN, Core Network, Transport). The corpus encodes **10 failure scenarios**, each comprising a causal chain spanning 2–5 files:

1. RRC Reconfiguration failure cascade (4 files, 13 steps)
2. SCTP transport link failure, 45 UEs affected (4 files)
3. Inter-gNB handover failure with PDCP data loss (4 files)
4. PDU session rejection by PCF policy (5 files)
5. RLC max retransmission causing DRB failure (4 files)
6. GTP-U path failure on backhaul (4 files)
7. 5G-AKA authentication failure, SQN desync (4 files)
8. DL scheduler starvation under load (3 files)
9. IPSec tunnel failure on midhaul (3 files)
10. AMF overload from registration storm (3 files)

Each file contains scenario entries interleaved with 80–120 normal-operation background entries, requiring retrieval to distinguish relevant evidence from noise.

Dataset Justification. The corpus is synthetic by design. Production telecom logs contain sensitive subscriber data, proprietary protocol traces, and operator-specific configurations that preclude public release under standard security and privacy constraints. A synthetic corpus offers three advantages for our study: (i) complete control over failure complexity and causal chain length, enabling systematic evaluation; (ii) known ground-truth root causes for every query, eliminating annotation ambiguity; (iii) full reproducibility, as both data and evaluation code are publicly released.

4.2 Queries

65 evaluation queries across 8 types: root-cause analysis (10), failure tracing (10), multi-hop (10), temporal (10), impact analysis (10), error-code (10), cross-scenario (4), contrastive (1). 55 of 65 require cross-file reasoning. Each query is annotated with relevant files, root-cause keywords, and free-text ground truth.

4.3 Ablation Configurations

Six configurations share the same LLM (Llama 3.3 70B):

Table 1: Ablation configurations.

Config	Dense	BM25	Rerank	Iter.
Dense-Only	✓			
BM25-Only		✓		
Hybrid	✓	✓		
Hybrid+Rerank	✓	✓	✓	
Hybrid+Iter.	✓	✓		✓
Full System	✓	✓	✓	✓

4.4 Multi-LLM Comparison

Retrieval is fixed (Hybrid+Rerank, top-6) across four models: Llama 3.3 70B, Llama 3.1 8B, Mixtral 8x7B, Gemma2 9B (all via Groq, temperature 0).

4.5 Metrics

Retrieval: Precision@K, Recall@K, MRR. **Reasoning:** keyword overlap (RCA_{kw}), semantic similarity, structured scoring, composite score ($0.35 \cdot \text{sem} + 0.30 \cdot \text{struct} + 0.20 \cdot \text{ROUGE-L} + 0.15 \cdot \text{kw}$). **LLM-as-judge:** binary correctness.

5 Results

5.1 Ablation Study

Table 2 reports aggregate metrics. The central finding: **retrieval metrics move; reasoning metrics do not**.

Table 2: Ablation results (65 queries, Llama 3.3 70B). Values produced by `run_ablation.py` and filled at run time.

Config	P@K	R@K	MRR	RCA_{kw}	Conf.	Iters
Dense-Only	0.553±0.18	0.500±0.19	0.613±0.18	0.311±0.16	0.553	1.0
BM25-Only	0.598±0.18	0.509±0.21	0.629±0.17	0.314±0.16	0.550	1.0
Hybrid	0.699±0.16	0.637±0.19	0.757±0.14	0.312±0.17	0.588	1.0
Hybrid+Rerank	0.794±0.15	0.720±0.17	0.822±0.14	0.344±0.16	0.641	1.0
Hybrid+Iter.	0.628±0.19	0.704±0.16	0.716±0.16	0.338±0.15	0.606	2.0
Full-System	0.751±0.15	0.729±0.15	0.811±0.14	0.359±0.18	0.649	2.2

Findings. (i) Cross-encoder reranking yields the largest single-component gain: Hybrid→Hybrid+Rerank improves P@K by +0.096 ($p < 0.001$), yet RCA_{kw} increases by only +0.032 ($p = 0.234$, n.s.). (ii) BM25 fusion adds complementary signal for exact error codes, lifting P@K from 0.553 (Dense-Only) to 0.699 (Hybrid). (iii) Iteration without reranking degrades precision via query drift (Hybrid+Iter. P@K=0.628 vs. Hybrid 0.699), though recall improves (0.704 vs. 0.637). (iv) Across all six configs, **P@K varies by 0.24** ($p < 0.001$) **while RCA_{kw} varies by only 0.05** ($p = 0.08$, n.s.)—the direct empirical signature of the retrieval–reasoning gap. Scatter plots of retrieval score vs. reasoning accuracy for all 390 query×config pairs are included in the supplementary material.

5.2 Multi-LLM Comparison

Table 3: Multi-LLM results with identical retrieval. Values from `run_multi_llm_comparison.py`.

Model	RCA_{kw}	Completeness	Latency	Params
Llama-3.3-70B	0.363±0.14	0.703±0.10	4.4	70B
Mixtral-8x7B	0.300±0.12	0.674±0.09	3.8	46.7B
Gemma2-9B	0.302±0.13	0.632±0.10	2.1	9B
Llama-3.1-8B	0.260±0.12	0.607±0.12	1.9	8B

With identical Hybrid+Rerank retrieval ($\text{P@K} \approx 0.794$ for all models), RCA_{kw} ranges from 0.260 (Llama 3.1 8B) to 0.363 (Llama 3.3 70B)—a 0.103 absolute gap attributable entirely to model capability. Completeness follows the same trend (0.607–0.703), confirming that **reasoning**

is independently bounded by model capability. Notably, the 70B→8B reasoning drop (0.103) exceeds the Dense-Only→Hybrid+Rerank retrieval-driven RCA change (0.032), indicating that model scaling dominates retrieval optimization for reasoning quality.

5.3 Retrieval–Reasoning Correlation

Table 4: Correlation between retrieval and reasoning across all runs. Values from `run_correlation_analysis.py`.

Metric	Pearson r	p	Spearman ρ	p
Precision@K	0.244	<0.001	0.222	<0.001
Recall@K	0.208	<0.001	0.178	0.0004
MRR	0.169	0.0008	0.143	0.0045

All correlations are weak ($|r|<0.25$, $|\rho|<0.23$), despite being statistically significant ($p<0.005$) due to the large sample ($n=390$). Pearson $r=0.244$ for P@K means retrieval quality explains only $\sim 6\%$ of variance in reasoning accuracy. This confirms the gap: **retrieval quality is a poor predictor of reasoning accuracy**.

5.4 Failure Mode Distribution

Table 5: Per-query regime classification.

Regime	Count	%
High retrieval, high reasoning	111	28.5
High retrieval, low reasoning	84	21.5
Low retrieval, high reasoning	84	21.5
Low retrieval, low reasoning	111	28.5

The 43% of queries in off-diagonal regimes (21.5% high-retrieval/low-reasoning + 21.5% low-retrieval/high-reasoning) are direct evidence that aggregate metrics obscure distinct failure modes. The symmetric distribution across quadrants further supports the independence claim: retrieval success neither guarantees nor precludes correct reasoning.

5.5 Statistical Significance

Table 6 reports paired t -tests (65 matched queries) with bootstrap 95% confidence intervals.

Table 6: Statistical significance of key comparisons (paired t -test, $n=65$).

Comparison	Metric	Δ	p	95% CI
Hybrid → Hybrid+Rerank	P@K	+0.096	<0.001***	[0.057, 0.134]
Hybrid → Hybrid+Rerank	RCA	+0.032	0.234	[−0.021, 0.085]
Dense-Only → Full System	P@K	+0.198	<0.001***	[0.149, 0.247]
Dense-Only → Full System	RCA	+0.048	0.083	[−0.006, 0.102]
Hybrid → Full System	P@K	+0.053	0.009**	[0.013, 0.092]
Hybrid → Full System	RCA	+0.047	0.132	[−0.014, 0.107]

Retrieval improvements are consistently significant ($p<0.01$), while reasoning differences are consistently non-significant ($p>0.08$). This pattern—significant retrieval gains paired with non-significant reasoning changes—is the statistical signature of the retrieval–reasoning gap.

5.6 Improved Metrics

Table 7: Metric inter-correlation. Values from `run_improved_metrics.py`.

Metric Pair	Pearson r
Keyword overlap vs. Semantic similarity	0.673
Keyword overlap vs. Composite score	0.773
Semantic sim. vs. ROUGE-L F1	0.628

Keyword overlap correlates only moderately with semantic similarity ($r=0.673$), indicating it is a noisy proxy—approximately 55% of variance is unexplained. The composite score correlates more strongly with semantic similarity ($r=0.849$), confirming that multi-signal evaluation reduces single-metric vulnerability.

5.7 Qualitative Cases

Case 1: Good retrieval, wrong reasoning. On SCTP failure (S02), Hybrid+Rerank retrieves all relevant files ($P@K > 0.8$), but the LLM stops at the symptom without tracing the full cascade.

Case 2: Imperfect retrieval, correct reasoning. On handover failure (S03), Dense-Only misses RAN entries, but the LLM correctly chains the failure using domain priors.

Case 3: Iteration helps. On authentication failure (S07), first-pass retrieval is broad; query refinement injects “SQN”, “AUSF”, pulling correct evidence on iteration 2.

Case 4: Iteration hurts. Under Hybrid+Iter. on scheduler starvation (S08), incorrect first-pass analysis causes query drift toward unrelated transport logs.

6 Discussion

6.1 Why Retrieval Improvements Fail to Improve Reasoning

The gap has a structural explanation: retrieval and reasoning are bounded by *different failure modes*. Retrieval is limited by index coverage, fusion weights, and reranker accuracy. Reasoning is limited by the LLM’s ability to chain multi-source evidence and distinguish relevant from irrelevant context.

Three mechanisms generate the disconnect: (i) Precision@K is reasoning-blind—a document can be relevant by annotation yet unused by the LLM. (ii) End-task accuracy is retrieval-blind—correct answers can arise from LLM priors rather than retrieved evidence. (iii) Averaging across queries erases the regime—two queries in opposite failure modes produce identical aggregate scores. The scatter plots in our supplementary materials (Figure 1) visualize this: across 390 query×config pairs, the regression slope is near-flat ($r=0.244$), with large vertical spread confirming that identical retrieval scores produce widely varying reasoning outcomes.

6.2 Implications for RAG Evaluation

Evaluating multi-hop RAG requires: (i) explicit decomposition of retrieval vs. reasoning failures; (ii) ablations holding the LLM fixed while varying retrieval, and vice versa; (iii) per-query regime reporting; (iv) semantic evaluation metrics beyond keyword overlap. These apply to multi-hop QA, scientific RAG, code assistants, and agentic retrieval.

6.3 Implications for Multimodal Systems

The retrieval–reasoning gap is likely amplified in multimodal settings. When a system retrieves heterogeneous evidence—images, tables, video segments, and text—from multiple sources, improving retrieval recall for each modality does not ensure the generation model can integrate cross-modal evidence into coherent reasoning. The challenge of assembling and reasoning over disparate evidence types mirrors the multi-hop text problem we study, but adds modality-alignment as an additional failure mode. Our decomposition framework (fixed retrieval $\times$ varied model, and vice versa) offers a principled evaluation template for multimodal generation pipelines facing this gap.

6.4 Adaptive vs. Fixed Iteration

Adaptive stopping avoids wasteful LLM calls and tolerates temporary confidence dips. Fixed iteration exhibits query drift (Case 4), amplifying first-pass errors without a precision-oriented reranking guard.

7 Limitations and Future Work

Text-only evaluation. All experiments use text-based retrieval and generation; extending to multimodal RAG (image+text, video+text) is necessary to confirm whether the retrieval–reasoning gap persists—or widens—across modalities.

Synthetic dataset. The corpus is synthetically generated with known ground truths. While this enables controlled evaluation, it may not capture the noise distributions, annotation ambiguity, and scale of production systems. Validation on real-world telecom logs and established multi-hop benchmarks (HotpotQA, MuSiQue) is needed to confirm generalizability.

LLM-as-judge bias. Cross-model judges and human evaluation are needed to validate automated scoring.

Single domain. Replication on additional domains (code-assistant, scientific QA, medical) would strengthen generality claims.

Confidence calibration. The composite confidence score is a heuristic, not a formally calibrated metric.

Future directions include: (i) multimodal retrieval–reasoning experiments with image+text evidence; (ii) larger corpora with bootstrap significance testing; (iii) human-calibrated evaluation; (iv) domain-adapted embeddings via contrastive learning; (v) graph-based causal retrieval; (vi) reader fine-tuning to test whether reasoning improvements close the gap.

8 Conclusion

We demonstrated the **retrieval–reasoning gap**: across six ablation configurations and 65 queries, P@K varies by 0.24 ($p<0.001$) while RCA_{kw} varies by only 0.05 ($p=0.08$, n.s.), with Pearson $r=0.244$ between retrieval and reasoning. A multi-LLM comparison with fixed retrieval shows reasoning accuracy gaps of up to 0.103 attributable to model capability alone. Per-query analysis reveals 43% of cases fall in off-diagonal regimes where retrieval and reasoning quality diverge. We proposed a decomposition framework and composite metric ($r=0.849$ with semantic evaluation) that make the gap visible. The broader implication: *for multi-hop RAG, improving retrieval is necessary but not sufficient—reasoning quality requires independent attention, and evaluation must reflect this.*

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